

Olist Customer Retention Analysis

Executive Summary

Prepared by Victoria Johnson | MySQL, Python, Power BI

Business Question

What separates one-time buyers from repeat customers, and what should Olist do about it?

The Finding

About 97% of customers placed only one observed order during the dataset period. Across 93,358 unique customers, 2,801 made more than one purchase. One-time and repeat customers had very similar average review scores at 4.15 and 4.21 respectively. Four of the top five product categories also appeared in both groups, although the ranking differed. Based on these comparisons, review score and product category alone do not appear to explain the low level of repeat purchasing.

Among customers who made a second purchase, the average time between the first and second order was 81.2 days. The analysis also found substantial differences in repeat purchase rates across sellers. These two patterns provide areas for further investigation and testing.

Why It Matters

Metric	One-Time Buyers	Repeat Customers
Customers	90,557 (97%)	2,801 (3%)
Total Revenue	R\$14.56M (94.4%)	R\$864K (5.6%)
Avg Observed Spend	R\$160.74	R\$308.53
Avg Review Score	4.15 / 5	4.21 / 5

Repeat customers spend 92% more per person than one-time buyers. Repeat customers spent an average of R\$308.53 per customer during the observed period, compared with R\$160.74 for one-time buyers. This is approximately 92% higher average observed spend.

Although repeat customers represented only a small share of the customer base, the difference in observed spend makes second-purchase behavior an important area for further investigation. Any revenue impact from a retention campaign should be measured through a controlled test rather than assumed from the historical spending difference.

Recommendation

1. Test a Day 55 - 65 Reactivation Campaign

Customers who made a second purchase returned after an average of 81.2 days.

Olist should test an automated re-engagement message between days 55 and 65 after a customer's first delivered order. The campaign could use personalized product recommendations or a limited incentive to encourage another purchase.

The 55 - 65 days period is a starting point for testing, not a proven optimal window. Results should be compared with a control group and measured using second-purchase conversion rate and incremental revenue.

2. Investigate Sellers with High Repeat Purchase Rates

Repeat purchase rates varied considerably across the sellers included in the analysis. The highest observed rate was 22.73%, compared with the platform average of 2.76%.

Olist should compare high-repeat sellers with similar sellers recording lower repeat rates. Delivery performance, pricing, promotions, product mix, and other available seller-level factors should be reviewed to identify differences associated with repeat purchasing.

3. Investigate Additional Causes of Low Repeat Purchasing

Similar average review scores and top category preferences suggest that these two factors alone do not explain the difference between one-time and repeat customers.

Further analysis should examine delivery times, late deliveries, cancellations, freight costs, payment behavior, and customer location. Customer acquisition source and post-purchase communication should also be reviewed if those data are available.

4. Measure Retention Using Customer Cohorts

The current analysis covers customers observed within a fixed dataset period. Customers who made their first purchase near the end of the dataset had less time to make another purchase.

Future retention analysis should group customers by first-purchase month and use a consistent observation window for each cohort. This would provide a clearer measure of repeat purchase behavior over time.

Method

Dataset: Olist Brazilian E-Commerce Public Dataset. 99,441 orders, 93,358 unique customers, and 8 source tables.

Tools: MySQL for analysis, Python with pandas and SQLAlchemy for database import and schema preparation, and Power BI for dashboard development.

Queries: Six SQL queries using CTEs, window functions including RANK, ROW_NUMBER, and LAG, multi-table joins, CASE WHEN segmentation, DATEDIFF for purchase-gap analysis, and conditional aggregation for seller-level repeat purchase rates.

Contact

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